

AMARNATH D.J.

Emerging AI Engineer

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SUMMARY

B.Tech Computer Science (AI & ML) student at SCTCE with hands-on experience building end-to-end machine learning pipelines, deploying production-grade web applications, and applying deep learning and computer vision techniques to real-world problems. Proficient in Python, Scikit-learn, TensorFlow, XGBoost, and Flask. Actively building expertise in data analysis, predictive modeling, and MLOps. Seeking roles in AI/ML Engineering, Data Science, Data Analytics, or Full-Stack Development.

SKILLS

- **AI & ML:** Scikit-learn, TensorFlow, Regression, Classification, Clustering, Model Evaluation Techniques
- **Deep Learning & Computer Vision:** YOLOv8, ByteTrack, Object Detection, Tracking, CNN, OpenCV
- **Data Science & Analysis:** Pandas, NumPy, Data Wrangling, EDA, Statistical Analysis, Data Cleaning
- **Data Visualization:** Matplotlib, Seaborn, Chart.js, Plotly
- **Languages:** Python, Java, C, SQL, HTML
- **Tools:** GitHub, Jupyter, VSCode, Notion
- **MLOps & Deployment:** Flask, FastAPI, Streamlit, Docker, Render, AWS (basic), Azure (basic)
- **Data Structures and Algorithms:** Arrays, Linked lists, stacks, queues, search and sort algorithms
- **Other Concepts:** Object Oriented Programming, Operating Systems, MS Word, MS Excel

PROJECTS AND RESEARCH

[GitHub](#)

- **Pre-Match Player Rating Prediction using Machine Learning**
Built a full-stack Machine Learning application using XGBoost and Flask to predict football player match ratings using engineered performance metrics such as rolling form, opponent strength, and rest days. Deployed the model as a production-ready web application.
- **Vision-Based Crowd Footfall Prediction System**
Developed a real-time crowd monitoring system using YOLOv8 and ByteTrack for detection and tracking, and applied SARIMA to forecast crowd flow from CCTV-derived time-series data.
- **Football Player Image Classification using Machine Learning**
Developed and deployed an AI-based football player recognition web application using feature-engineered ML (Wavelet + PCA + Logistic Regression) with Flask on Render.
- **Bangalore Home Prices – Real Estate Price Prediction System**
Built an ML-based web application to predict house prices using regression models. Developed Flask APIs for real-time predictions and integrated with a responsive frontend.
- **Machine Learning Foundation Projects (10+ core concept projects):**
Implemented multiple ML algorithms (Linear Regression, Logistic Regression, SVM, Random Forest) from scratch and applied them to real-world datasets including housing price prediction and image classification.
- **Data Visualization projects (in progress):**
Developing a series of sports analytics visualizations using Python, Pandas, FastAPI, and Chart.js, building interactive dashboards to analyze metrics such as goal scoring and player creativity.

CERTIFICATIONS

- Certificate of Completion (NPTEL): Python for Data Science – *IIT Madras*
- Certificate of Completion (LEAP): Reverse Engineering Bootcamp – *IIT Madras*
- Certificate of Completion (LEAP): Automation with Arduino Bootcamp – *IIT Madras*

EDUCATION

- Loyola School, Thiruvananthapuram – ISCE/ISC Board {2008–2022}
- Sree Chitra Thirunal College of Engineering, Thiruvananthapuram – B.Tech Degree – AI/ML {2023–Present}

ACTIVITIES AND ACHIEVEMENTS

- KTU A-Zone Chess Championship 2025 Runners-Up
- SCTCE Inter department Chess Championship Winner 2025
- SCTCE Chess Club Male Lead 2025/26